

Week 5

Monday 2/9

ex | Derive an explicit two step LMM

$$w_{i+1} - a_0 w_i - a_1 w_{i-1} - h(b_0 f_i + b_1 f_{i-1}) = 0$$

	y	y'
1	1	0
0	$-a_0$	$-b_0$
-1	$-a_1$	$-b_1$

	1	0
1	1	0
1	0	0
1	0	0

	x	1
1	1	1
0	0	1
-1	1	1

	x^2	$2x$
1	1	2
0	0	0
1	1	-2

	x^3	$3x^2$
1	1	3
0	0	0
-1	1	3

$$1 - (a_0 + a_1) = 0 \quad 1 + a_1 - b_0 - b_1 = 0 \quad 1 - a_1 + 2b_1 = 0 \quad 1 + a_1 - 3b_1 = 0$$

$$a_0 = -4, \quad a_1 = 5 \quad b_0 = 0 \quad b_1 = 2$$

$$w_{i+1} + 4w_i - 5w_{i-1} - 2hf_{i-1} = 0$$

recall: why are we so wrong?

→ using a difference equation to approx an ODE

1) consistency (✓)

2) convergence (✗)

3) numerical stability (?)

→ recall: 2nd order ODE:

$$y'' + ay' + by = 0$$

→ solve homogeneous problem → $y = e^{rt}$ → gives characteristic polynomial

→ solve nonhomogeneous problem

note: theory to solving difference eq. is nearly identical to ODEs

→ consider $y' = 0, y(0) = 0$

→ gives homogeneous difference eqn:

$$w_{i+1} + 4w_i - 5w_{i-1} = 0, \quad \text{assumes soln. is of the form } w_i = c\lambda^i$$

sub in $\{ c\lambda^{i+1} + 4c\lambda^i - 5c\lambda^{i-1} = 0$

$$c(\lambda^2 + 4\lambda - 5)\lambda^{i-1} = 0 \quad \rightarrow \quad w_i = c_1(1)^i + c_2(-5)^i \rightarrow \text{apply initial cond}$$

$\lambda = 1, -5$ $c_1 + c_2 = 0$ $c_1 = -c_2$

$$w_i = c_i (1^i - (-5)^i) > 0$$

$c_1 = 0 \rightarrow$ computer will "think" $0 +$ roundoff error

$-5^i \leftarrow$ this is the **problem** & leads to exp. growth due to roundoff $\rightarrow$ **Blow up**

$\Rightarrow$ if you have a consistent, stable method, rest will follow

Root condition: if roots λ of the **characteristic** polynomial are st $|\lambda| \leq 1$ & roots with $|\lambda| = 1$ polynomial $\rightarrow$ are simple, then the difference equation satisfies the **root condition** growth/nonrepeating

Back to our 2 step method...

$$1 - a_0 - a_1 = 0 \quad \left. \begin{array}{l} \text{char. poly: assume } \delta = 0 \\ w_{i+1} - a_0 w_i - a_1 w_{i-1} = 0 \end{array} \right\}$$

$$1 + a_1 - b_0 - b_1 = 0 \quad \left. \begin{array}{l} \lambda^{i-1} (\lambda^2 - a_0 \lambda - a_1) = 0 \end{array} \right\}$$

$$1 - a_1 + 2b_1 = 0 \quad \rightarrow \lambda^2 - a_0 \lambda - (1 - a_0) = 0$$

$$1 + a_1 - 3b_1 = 0 \quad \rightarrow \lambda^2 - a_0 \lambda - (1 - a_0) = 0$$

$$(\lambda - 1)(\lambda - (a_0 - 1)) = 0$$

$$a_0 - 1 = \lambda \Rightarrow -1 < a_0 - 1 \leq 1 \Rightarrow 0 \leq a_0 \leq 2 \Rightarrow \text{repeating behavior @ edge } 0 < a_0 < 2$$

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$$(\lambda - 1)(\lambda - (a_0 - 1)) = 0$$

$$\rightarrow \text{want } |\lambda| \leq 1 \rightarrow 0 \leq a_0 \leq 2$$

$$\rightarrow \text{recall: } w_i = (\alpha + \varepsilon_m)(1^i + \lambda^i)$$

more generally:

$$w_{i+1} = a_0 w_i + a_1 w_{i-1} + \dots + a_{s-1} w_{i-(s-1)}$$

guess: $w_i = \lambda^i$; consider $y' = 0$, $y(0) = \alpha$ } trivial to tell us about stability

$$\lambda^{i-(s-1)} [\lambda^s - a_0 \lambda^{s-1} - a_1 \lambda^{s-2} - \dots - a_{s-1}] = 0$$

Since $y(0) = \alpha$, then true soln is $w_i = \alpha$

sub in $w_i = \alpha$

$$\alpha [1 - a_0 - a_1 - a_2 - \dots - a_{s-1}] = 0$$

using these 2 eqn, it

shows us that $\lambda = 1$ is ALWAYS a root

if roots are **distinct**, then the general soln has the form

$$w_i = \sum_{k=1}^s c_k \lambda_k^i \quad (\text{also all } |\text{roots} - 1| < 1)$$

if roots are **repeated**, then the general soln includes these terms...

$$c_k \lambda_k^i + c_{k+1} i \lambda_k^i + c_{k+2} i^2 \lambda_k^i + \dots + c_{k+\beta} i^\beta \lambda_k^i$$

repeated root

if $|\lambda| = 1$, there is polynomial growth $\rightarrow$ numerical instability $\gg$

otherwise, if $|\lambda| < 1$, this will be a stable method

Root End & Numerical stability:

if $|\lambda_i| \leq 1$, $i = 1, 2, \dots, s$; then, the s-step diff method satisfies the root condition; all roots st $|\lambda_i| = 1$ are simple

Dahlquist stability Theorem:

A LMM is convergent iff it is stable and consistent

stability: a method is strongly stable if only one root has $|\lambda| = 1$ and remaining roots are st $|\lambda| < 1$

- a method is weakly stable if it has more than one root w/ $|\lambda| = 1$
- unstable otherwise

I thought if there is more than 1 root w/ $|\lambda| = 1 \rightarrow$ repeated

what is the difference between multiple simple roots = 1 vs repeated roots?

Is it that if $\lambda_k = \lambda_{k+1}$ it is repeated but not if $\lambda_k \neq \lambda_{k+1}$ but $\lambda_k = \lambda_{k+2}$?

Adam Bashforth methods (explicit)

Since char. poly $p(\lambda)$ has a root @ $\lambda = 1$, then "most" stable method

$$\rightarrow w_{i+1} = a_0 w_i + h \phi_i$$

$$\rightarrow \text{char poly: } (\lambda - 1) = 0$$

Friday 2/3

most stable LMM - Adams Bashforth & Adams - Moulton:

char poly: $\lambda(\lambda-1)=0$

ex) AB 2-step

$$\begin{array}{c} y \quad y' \\ \begin{array}{|cc|} \hline 1 & 0 \\ 0 & -a_0 - b_0 \\ -1 & 0 - b_1 \\ \hline \end{array} \end{array}$$

AB 3

$$\begin{array}{c} y \quad y' \\ \begin{array}{|cc|} \hline 1 & 0 \\ 0 & -a_0 - b_0 \\ -1 & 0 - b_1 \\ -2 & 0 - b_2 \\ \hline \end{array} \end{array}$$

$$\begin{array}{c} \begin{array}{|cc|} \hline 1 & 0 \\ \hline 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ \hline \end{array} & \begin{array}{c} x \quad 1 \\ \begin{array}{|ccc|} \hline 1 & 1 & 1 \\ 0 & 0 & 1 \\ -1 & -1 & 1 \\ \hline \end{array} \end{array} & \begin{array}{c} x^2 \quad 2x \\ \begin{array}{|cc|} \hline 1 & 2 \\ 0 & 0 \\ -1 & -2 \\ \hline \end{array} \end{array} & \begin{array}{c} x^3 \quad 3x^2 \\ \begin{array}{|cc|} \hline 1 & 3 \\ 0 & 0 \\ -1 & -3 \\ \hline \end{array} \end{array} \end{array}$$

$$a_0=1 \quad 1-b_0-b_1=0$$

$$1+2b_1=0$$

$$1-3b_1=0$$

$$b_0=3/2$$

$$b_1=-1/2$$

$$a_0=1$$

$$1+\frac{3}{2}=\frac{5}{2} \leftarrow \text{coeff for local error}$$

$$\hookrightarrow \frac{5}{2}h^3 = \text{local error}$$

$$\therefore O(h^2) \text{ for global error}$$

AB3

	y	y'	1	0	x	1	x ²	2x	x ³	3x ²	x ⁴	4x ³
1	1	0	1	1	0	1	1	2	1	3	1	4
0	-a ₀	-b ₀	0	1	0	0	0	0	0	0	0	0
-1	0	-b ₁	-1	1	0	-1	-1	-2	-1	-3	1	-4
-2	0	-b ₂	-2	1	0	-2	-2	-4	-2	-8	16	-32

$$\begin{aligned}
 1 - a_0 &= 0 & 1 - b_0 - b_1 - b_2 &= 0 & 1 + 2b_1 + 4b_2 &\stackrel{!}{=} 0 & 1 - 3b_1 - 12b_2 &= 0 & 1 + 4b_1 + 32b_2 &= 0 \\
 a_0 &= 1 & b_0 &= 1 - b_1 - b_2 & 3(2b_1 + 4b_2) &= -1 & -\frac{8}{3} + 4b_2 &= -1 & \frac{3}{3} + \frac{-16}{3} + \frac{8}{3} &= -5/3 h^4 \\
 & & & & -3b_1 - 12b_2 &= -1 & 4b_2 &= \frac{5}{3} & & \\
 & & & & 3b_1 &= -4 & b_2 &= 5/12 & & \\
 b_0 &= \frac{12 + 16 - 5}{12} = \frac{23}{12} & & & b_1 &= -4/3 = -16/12 & & & & \frac{-5}{3(4!)} = \frac{-5}{72} \\
 & & & & & & & & & \frac{3}{8} h^3 (y^{(4)})
 \end{aligned}$$

$$w_{i+1} = w_i + \frac{h}{12} (23f_i - 16f_{i-1} + 5f_{i-2}) \quad O(h^3)$$

AB4

1	0
-1	-55/24
0	59/24
0	-37/24
0	9/24

global error:
25/720 h⁴ y⁽⁵⁾

Adams-Moulten:

AM2

$$w_{i+1} = a_0 w_i + h(b_{-1} f_{i+1} + b_0 f_i + b_1 f_{i-1})$$

	y	f	1	0	x	1
i+1	1	-b ₋₁	1	0	1	1
i	-a ₀	-b ₀	1	0	0	1
i-1	0	-b ₁	0	0	-1	1

AM2

1	5/12
-1	2/3
0	-1/12

global error:
 $-\frac{1}{12} h^3 y^{(4)}$

We want to compare via # of function eval $\therefore$ compare AM2 & AB2

$$\rightarrow \left| -\frac{1}{12} \right| < \left| \frac{3}{8} \right|$$

↳ how to deal w/ this

$x = g(x) \rightarrow$ deal w/ iterations (Newton's Method) FPI

"Stiff" ODEs

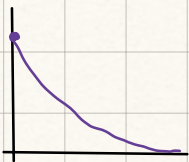
→ "define" & numerical issues

convergence $\Leftrightarrow$ stability (num) & consistency

↳ $\lim_{h \rightarrow 0} |y_i - w_i| = 0$, in practice $h \neq 0$ & h is fixed

ex | Consider $y' = -10y$ $y(0) = 1$

↳ $y = e^{-10t}$

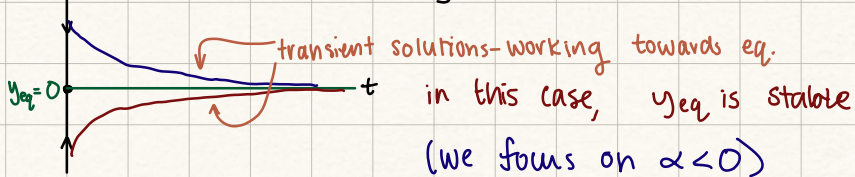


Increasing $h \rightarrow$ large oscillations & does not decay to 0

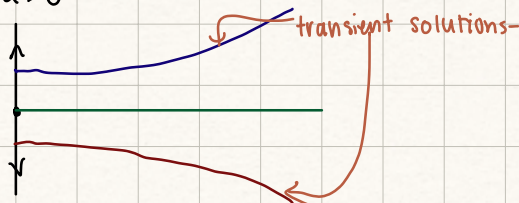
recall: two solution types:

- 1) equilibrium solution
- 2) transient solutions

phase lines: concerned w/ $y' = \alpha y$, $\alpha < 0$



if $\alpha > 0$



↓
make sure we
have stable
asymptotic behavior